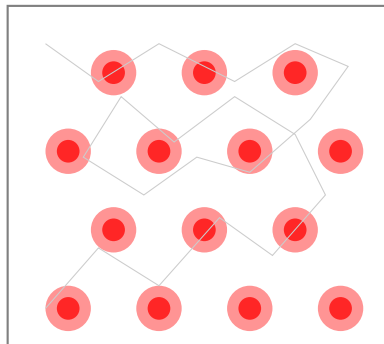


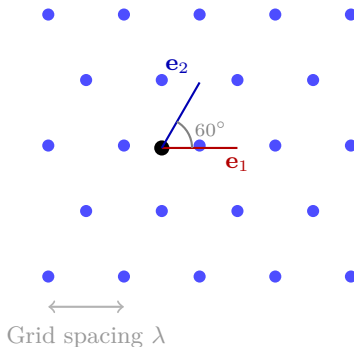
Grid Cell Hexagonal Firing Pattern

(A) Spatial firing fields



Position in arena

(B) Hexagonal lattice structure



(C) Phase wrapping produces toroidal topology

$$\begin{array}{c}
 \lambda \qquad \lambda \\
 \text{Spatial position } x \\
 \hline
 \end{array}
 \quad \xrightarrow{\text{mod } \lambda} \quad
 \begin{array}{c}
 \text{Blue circle} \\
 \phi_1 \in S^1
 \end{array}
 \times
 \begin{array}{c}
 \text{Green circle} \\
 \phi_2 \in S^1
 \end{array}
 =
 \begin{array}{c}
 \text{Pink torus} \\
 T^2 = S^1 \times S^1
 \end{array}$$

Periodic spatial coding in two directions produces two cyclic phase variables. Their joint state space is the two-torus T^2 .